

Arjun Patel

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EDUCATION

University of California, Berkeley | GPA: 3.90/4.00 | Berkeley, CA Dec. 2026
Double Major: Astrophysics & Applied Mathematics Concentrating in Applied Discrete Probability
Brighton High School | GPA: 3.98/4.00 | ACT: 36 | Rochester, NY Jun. 2023

PROFESSIONAL EXPERIENCE

Quantitative Analyst Intern *AIN Ventures* | New York, NY May. 2025 - Present

- Developing a Monte Carlo simulation framework to model venture portfolio outcomes under uncertainty.
- Collaborating with MP to integrate AI-driven probabilistic modeling into internal forecasting systems.

Research Assistant *Rubin Observatory* | Berkeley, CA Aug. 2024 - Present

- Engineered a pipeline to evaluate ≈ 8 months of telescope chamber temperature data to assess thermal stability.
- Analyzed images from 9 sensors, generating heat-map visualizations to evaluate atmospheric distortion effects.
- Performed Fourier transforms on star profiles using 3 averaging techniques to detect spectral corruption.
- Collaborating with Berkeley Dean Steven Kahn and Stanford PhD candidate on data interpretation.

Quantitative Trading Intern *Flow Traders Ascent* | New York, NY Apr. 2025 - Jun. 2025

- Participated in trading bot competition, placing 2nd among interns using market-making & arbitrage strategies.
- Exposed to trading workflow through meetings, trader Q&A sessions, and shadowing with Flow's trading team.

Fellow *NASA Ames Research Center* | Berkeley, CA Sep. 2024 - May. 2025

- Identified top 3 technology gaps across 7 critical areas for small satellite advancements for NASA missions.
- Analyzed data from 50+ government, commercial, & academia organizations; conducted 10+ expert interviews.
- Developed strategic frameworks to monitor and address technology gaps for future exploration missions.

Data Analyst *NASA SPORES - HWO* | Berkeley, CA Jan. 2025 - May. 2025

- Processed 10K+ radial velocity data points to update exoplanet mass and orbit constraints.
- Identified and standardized datasets from major spectrographs like HIRES/Keck and HARPS/ESO.
- Published in an upcoming research paper for contributions to exoplanet detection research.

Data Science & Engineering Intern *Dotlas* | Berkeley, CA Oct. 2023 - Jan. 2025

- Deployed 10+ robust Python scrapers covering 10M+ records across hospitality, retail, and traffic data sources.
- Designed official data visualization website using Streamlit to enhance internal data accessibility.
- Built a pipeline to automatically extract social URLs with 90% accuracy, streamlining data collection.
- Created LLM - based prompts and agents to return organized promotional content and event details.
- Indexed 5000+ brands using custom ranking algorithms to benchmark presence and identify top 20 competitors.

EXTRACURRICULAR ACTIVITIES

Telescope Officer *Undergraduate Astronomy Society* | Berkeley, CA Aug. 2023 - Present

- Guided 10+ members in operating a variety of telescopes, including Treffers, Cassegrain, Newtonian, and EV.
- Managed telescope operations for 8+ events per semester, ensuring precise alignment and optimal performance.

Competitor *Poker @ Berkeley* | Berkeley, CA Sep. 2023 - Present

- Placed 3rd in Jane Street and 2nd in Flow Traders Estimathon Contests, solving real-world estimation problems.
- Compete in weekly poker sessions, club-hosted events, Intro to Poker class, and 5+ tournaments across California.

PROJECTS

Adaptive ML - Based Momentum & Mean Reversion Trading

- Trained machine learning model to detect trends & switch between momentum and mean-reversion strategies.
- Engineered features like rolling volatility, z-scores, and stop loss risk to optimize signal accuracy.
- Outperformed the S&P 500 by 3 $\times$ over the past five years through dynamic strategy selection.

COURSEWORK, SKILLS & INTERESTS

Coursework: Quantum Mechanics, Probability & Random Processes, Real Analysis, Abstract Algebra, Stellar Physics, Combinatorics, Discrete Math, Linear Algebra & Differential Equations, Numerical Analysis, Quant Finance, Akuna 101
Skills: Python, SQL, Matlab, Java, GitHub, Jupyter, Databricks, AWS, GCP, Git, Docker, LLMs, Microsoft Office
Interests: Astronomy, Poker, Math, Finance, Lifting, Gaming, Football, Marvel, Legos, Board Games, Cooking